



CORRELATION

Housekeeping

1. Class Recording Permission

One of our participants is affected by the current shutdown and cannot attend sessions. To support him, I'd like to record today's class.

If you prefer not to appear in the video, you're welcome to turn off your camera.

Housekeeping

2. Your final assignment is out

Deadline is on the 29th of October.

An additional data source idea, if you know geographical data and like working with it <https://places.foursquare.com/>

- What did you learn from our last lecture?
- Can you think of an example from your own work or research where you used scatterplots to tell a data story?

"Personally, I prefer ordinarily to stick within the range extending from a ratio of 1:1.5 to 1:0.5 and lean toward a ratio of 1:1 as ideal".

at the next five figures and determine which seems to work best:

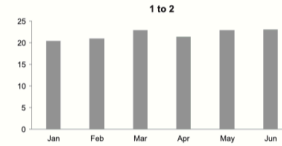
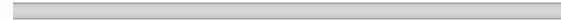


FIGURE 10.13 This is an example of a 1-to-2 ratio of bar width to intervening white space.



COMPONENT-LEVEL GRAPH DESIGN 213



FIGURE 10.14 This is an example of a 1-to-1.5 ratio of bar width to intervening white space.



FIGURE 10.15 This is an example of a 1-to-1 ratio of bar width to intervening white space.



FIGURE 10.16 This is an example of a 1-to-0.5 ratio of bar width to intervening white space.

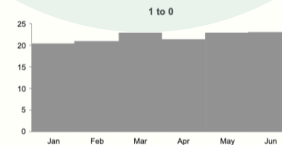


FIGURE 10.17 This is an example of a 1-to-0 ratio of bar width to intervening white space.

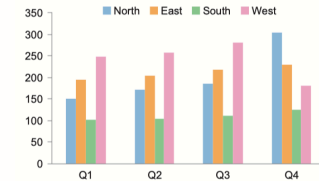


FIGURE 10.18 This graph illustrates when it is appropriate to place bars side by side without white space in between.

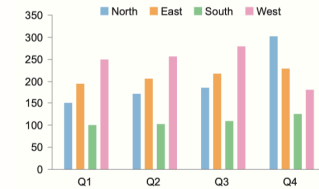
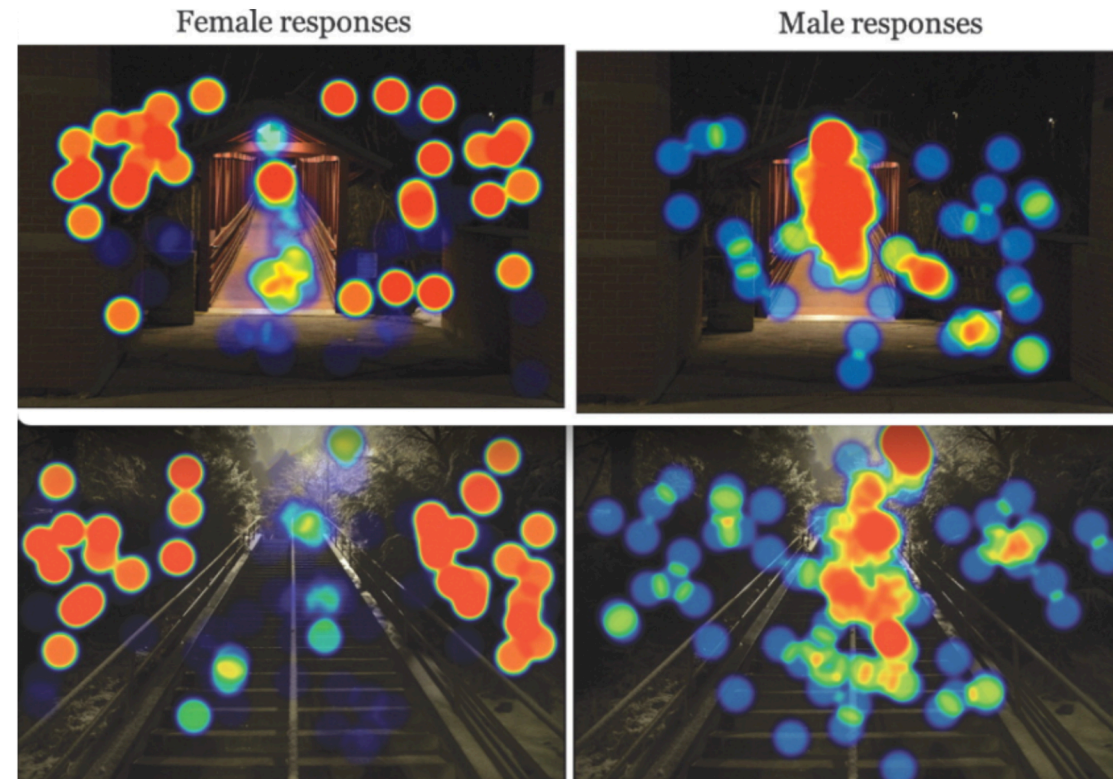


FIGURE 10.19 This graph shows that it's unnecessary to insert white space between bars that are associated with the same categorical item along the axis.

HEATMAP SCATTERPLOTS

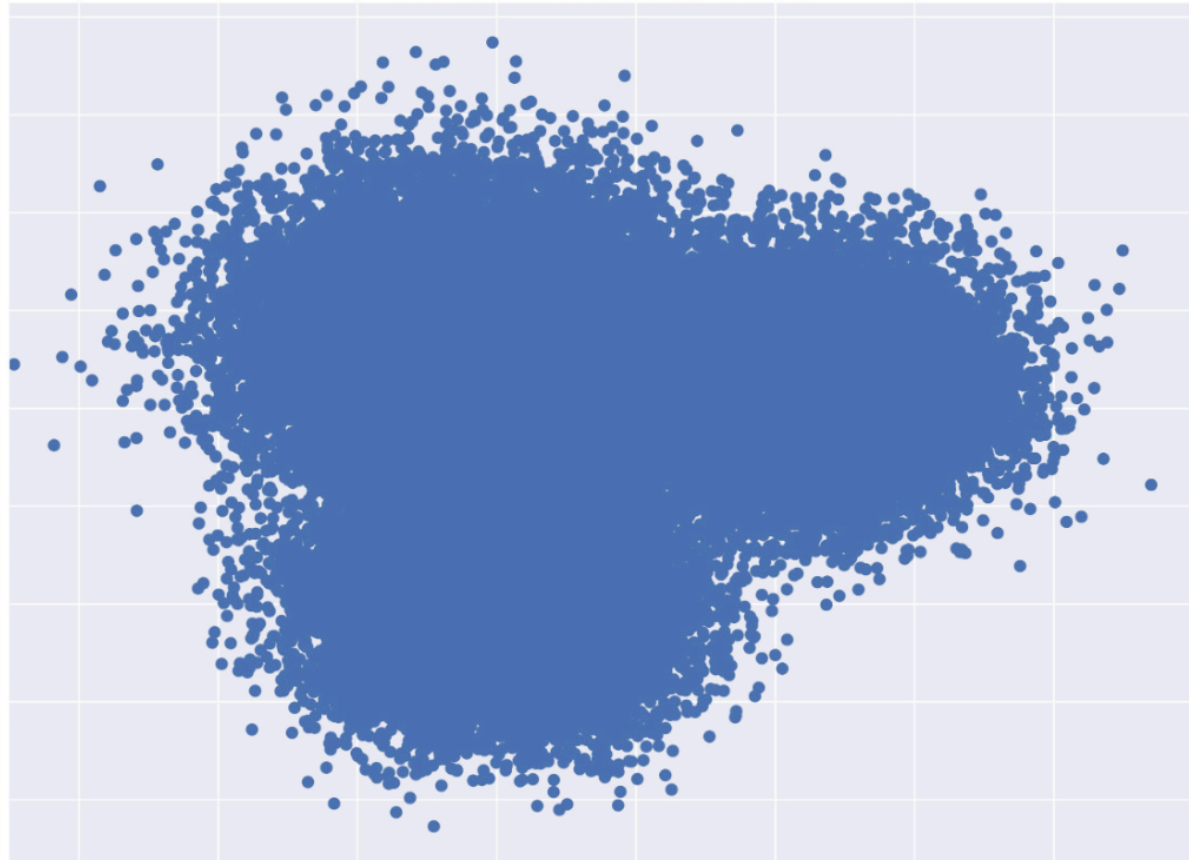
Heat maps show men look straight ahead; women scan periphery



An eye-catching new BYU study shows just how different the experience of walking home at night is for women versus men.

<https://www.liebertpub.com/doi/full/10.1089/vio.2023.0027>

CORRELATIONS - SCATTERPLOTS

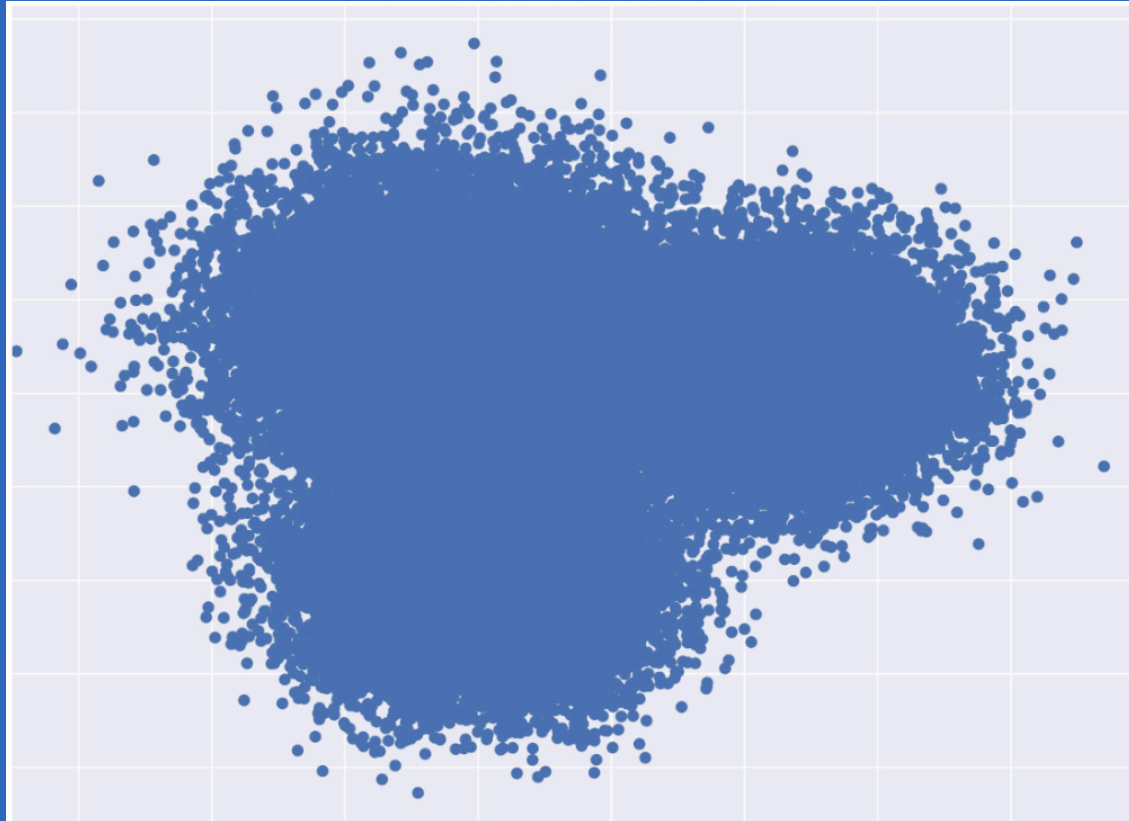


What is a scatterplot?

WHAT IS A SCATTERPLOT?

- A scatterplot is a graph that is used to display the relationship between two quantitative variables and is described by the direction, strength, and form of the relationship, and unusual observations.
- Direction: A positive direction means that the y-variable tends to increase as the x-variable increases (or an upward trend). A negative direction means that the y-variable tends to decrease as the x-variable increases (or a downward trend). Relationships between quantitative variables are said to have no direction when there is no consistent positive or negative pattern in the scatterplot.
- Strength: Relationships between two quantitative variables run from strong, if the points fall in a “tight” pattern, to weak, if they are spread out.
- Form: The form of a relationship is determined by the pattern of the points in the scatterplot. The most common forms are linear and curved. If there is no clear pattern in the scatterplot, the form is sometimes referred to as scattered.
- Unusual observations: Unusual points that do not fit the general pattern in the scatterplot may be outliers.

CORRELATIONS - SCATTERPLOTS



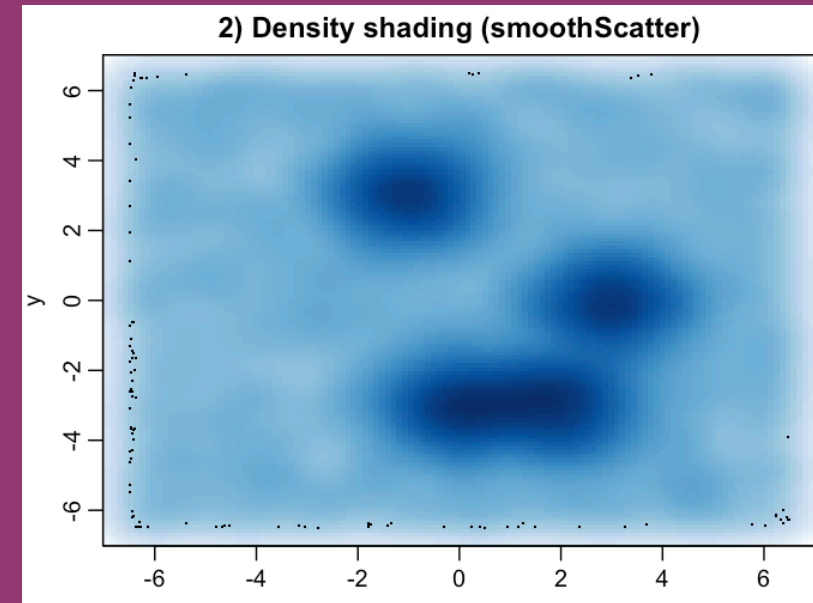
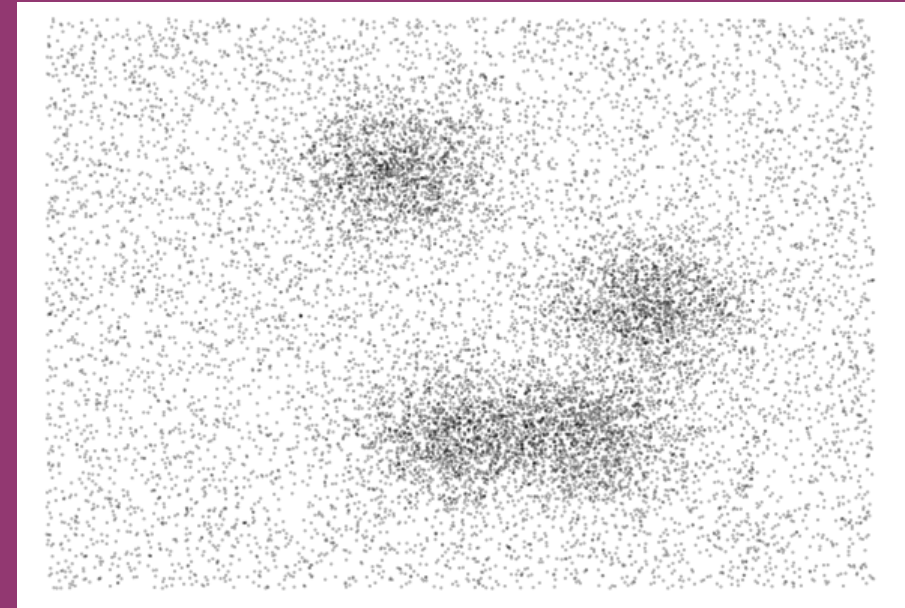
- Decrease symbol size
- Remove fill color
- Try different symbols
- Add random noise (jittering)
- Use alpha parameter
- Use subsamples

What can we do to make our scatterplots more readable?

WHAT CAN WE DO TO MAKE OUR SCATTERPLOTS MORE READABLE?

- Create a density plot

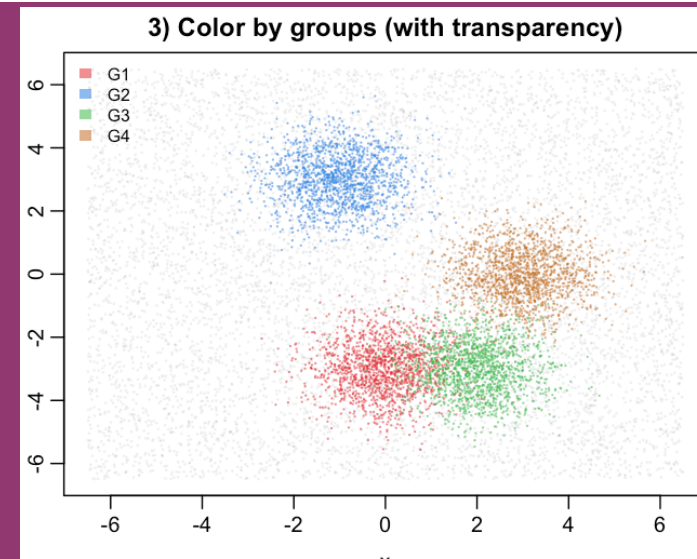
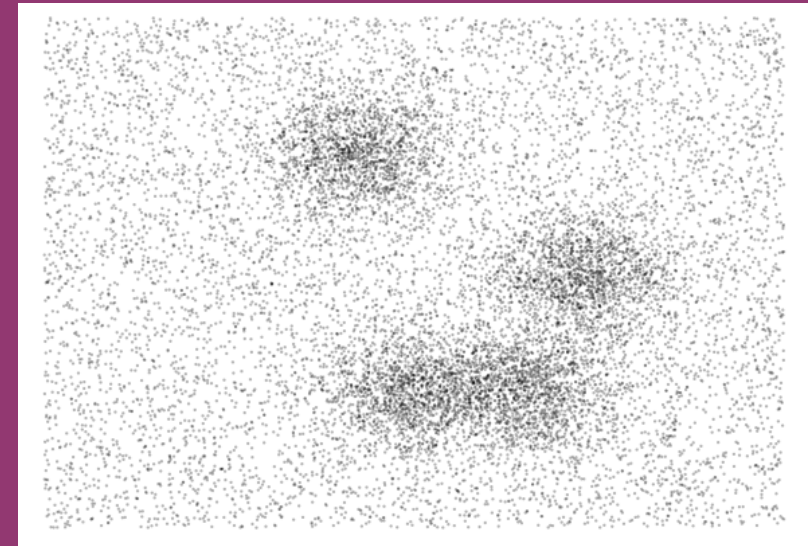
```
smoothScatter(dat$x, dat$y, xlab = "x", ylab = "y",  
main = "2) Density shading (smoothScatter)", xlim =  
xlim, ylim = ylim)
```



WHAT CAN WE DO TO MAKE OUR SCATTERPLOTS MORE READABLE?

- Colour by a known group

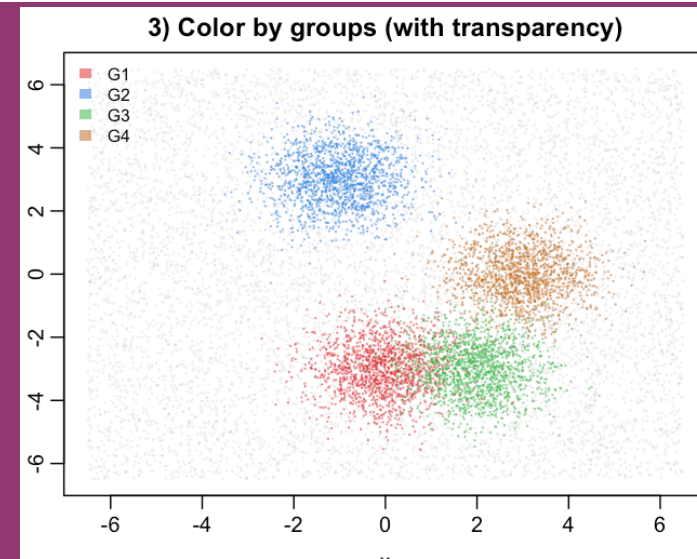
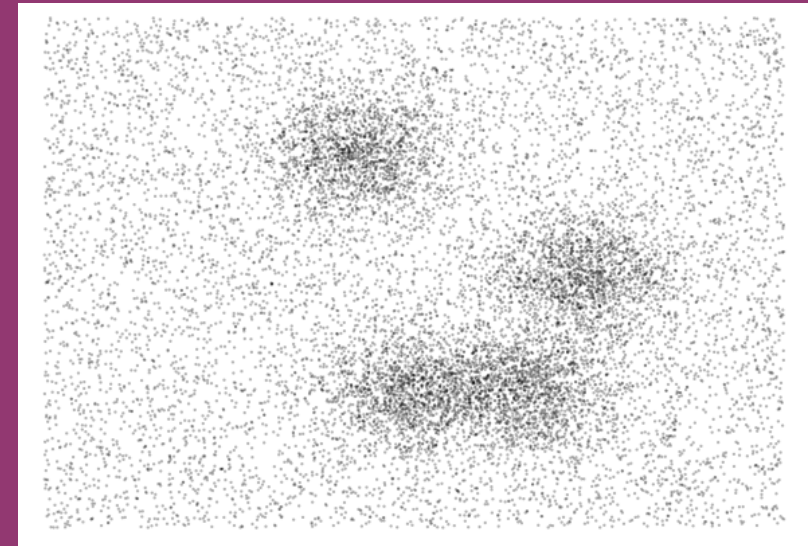
```
legend("topleft", fill = pal[c("G1","G2","G3","G4")],  
border = NA, legend = c("G1","G2","G3","G4"), cex =  
0.8, bty = "n")
```



WHAT CAN WE DO TO MAKE OUR SCATTERPLOTS MORE READABLE?

- Colour by a known group

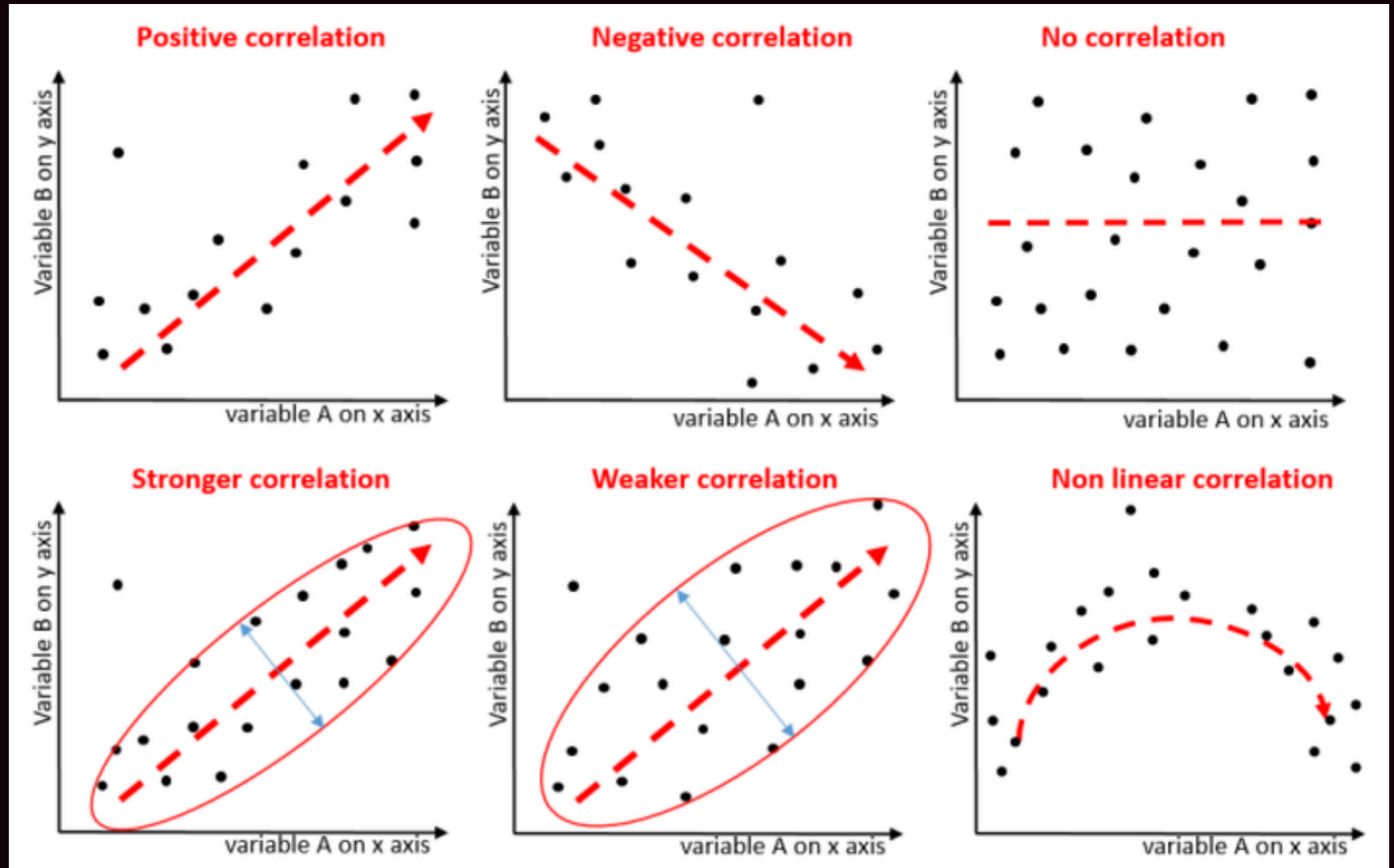
```
legend("topleft", fill = pal[c("G1","G2","G3","G4")],  
border = NA, legend = c("G1","G2","G3","G4"), cex =  
0.8, bty = "n")
```



SCATTERPLOTS

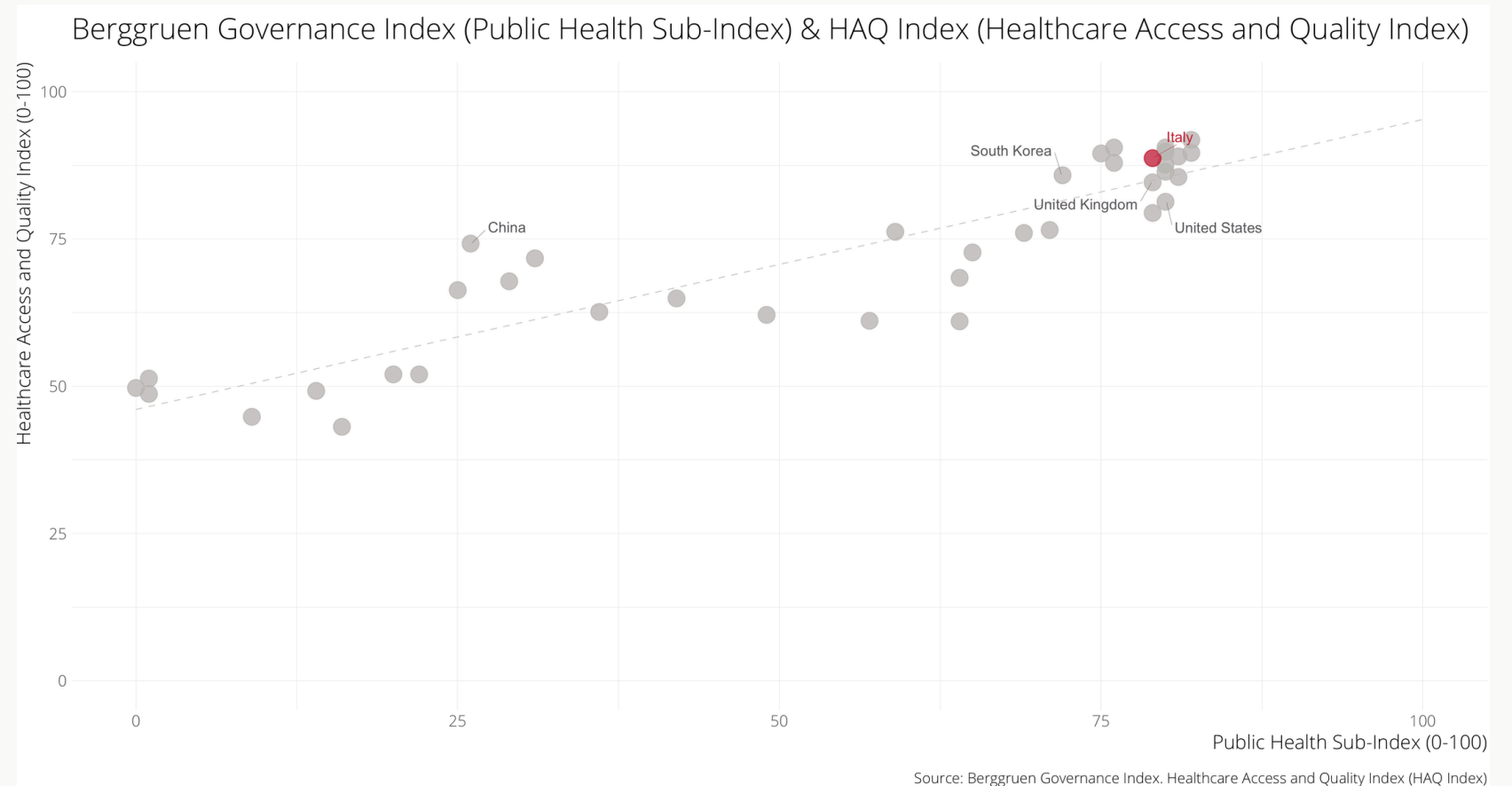
- Scatter plots can encode up to five different parameters:
 - X and Y axes
 - Color
 - Size
 - Shape of markers
- Readers may have difficulty perceiving > 3 parameters simultaneously
- Scatter plots are less useful when data are very dispersed. It is better when we can see a trend or some grouping

TREND: CORRELATIONS



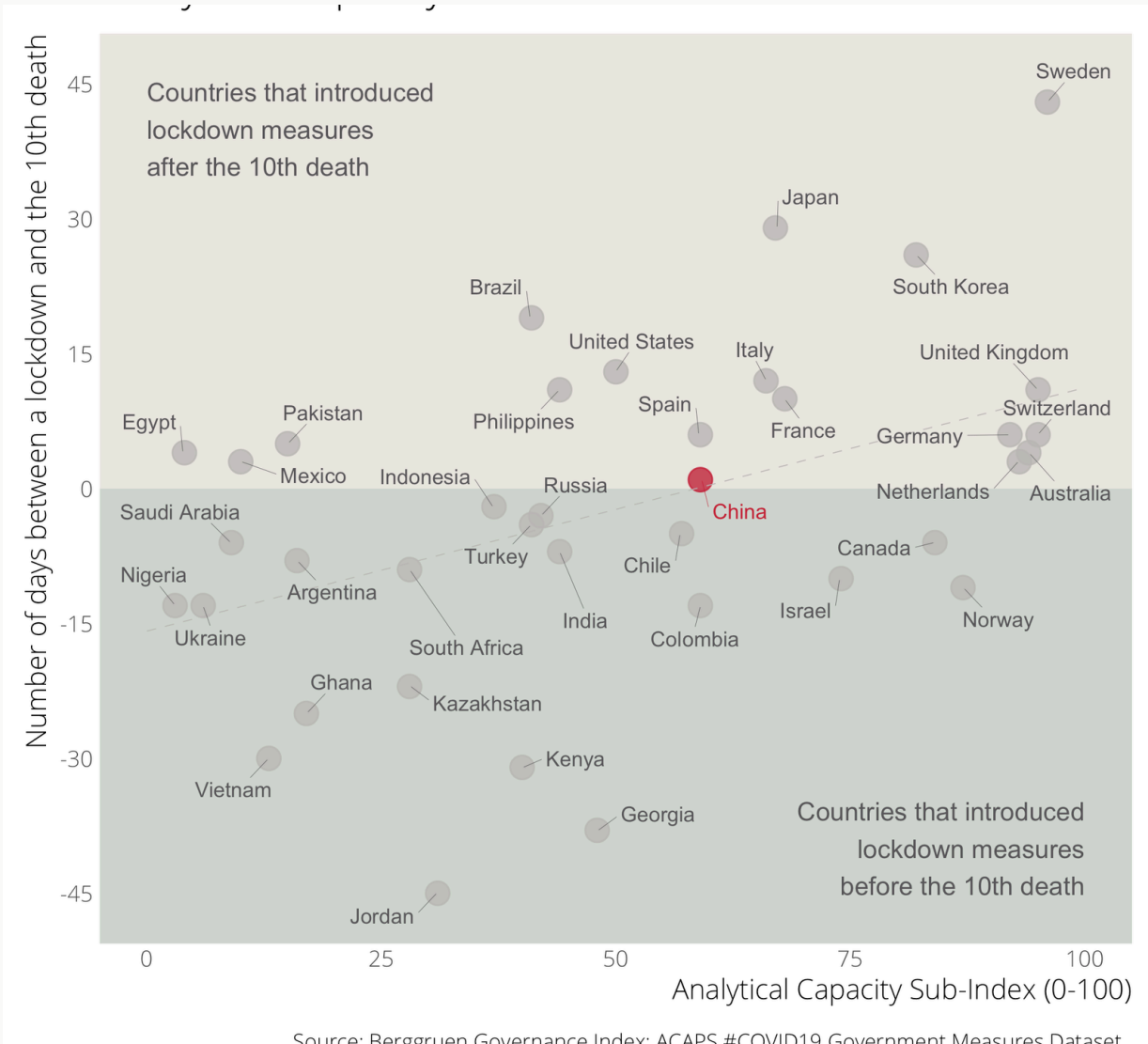
TREND

(based on the data points)



TREND

(based on the data points)

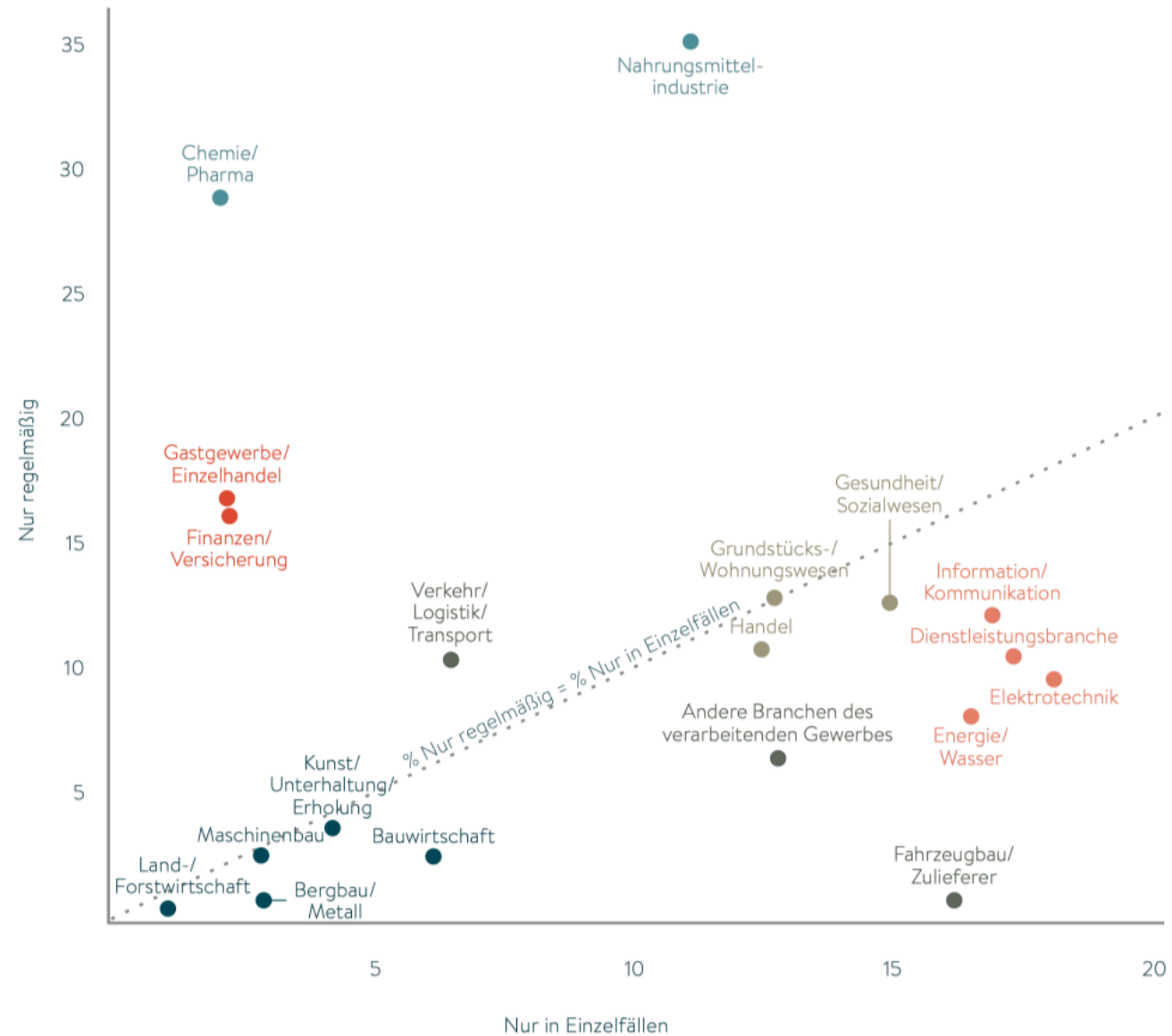


How would you improve this chart?

TREND

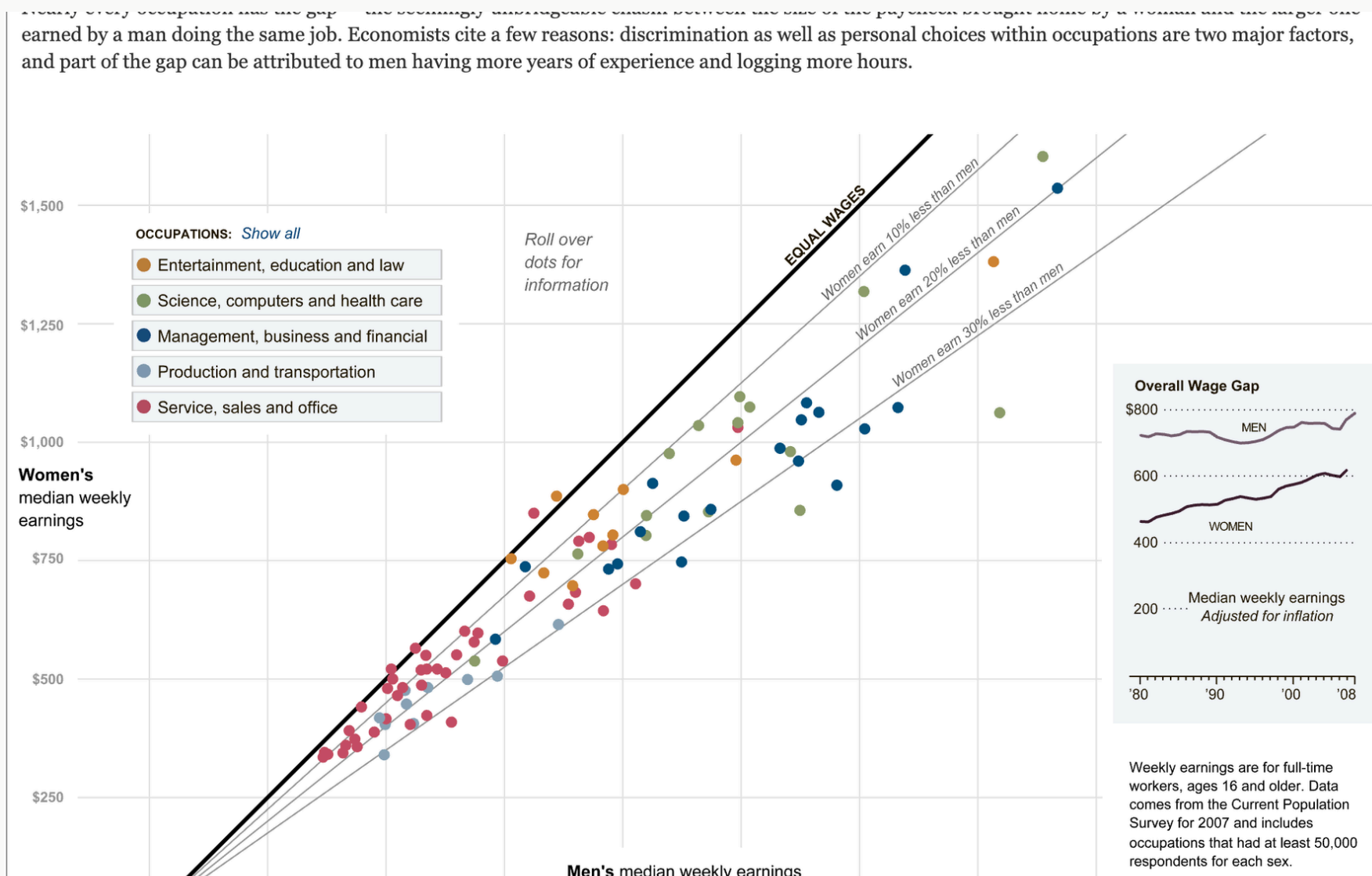
based on the axis
"equality"

Prozent der Unternehmen, die in der jeweiligen Branche angeben, regelmäßig oder in Einzelfällen mit INGOs zusammenzuarbeiten



TREND

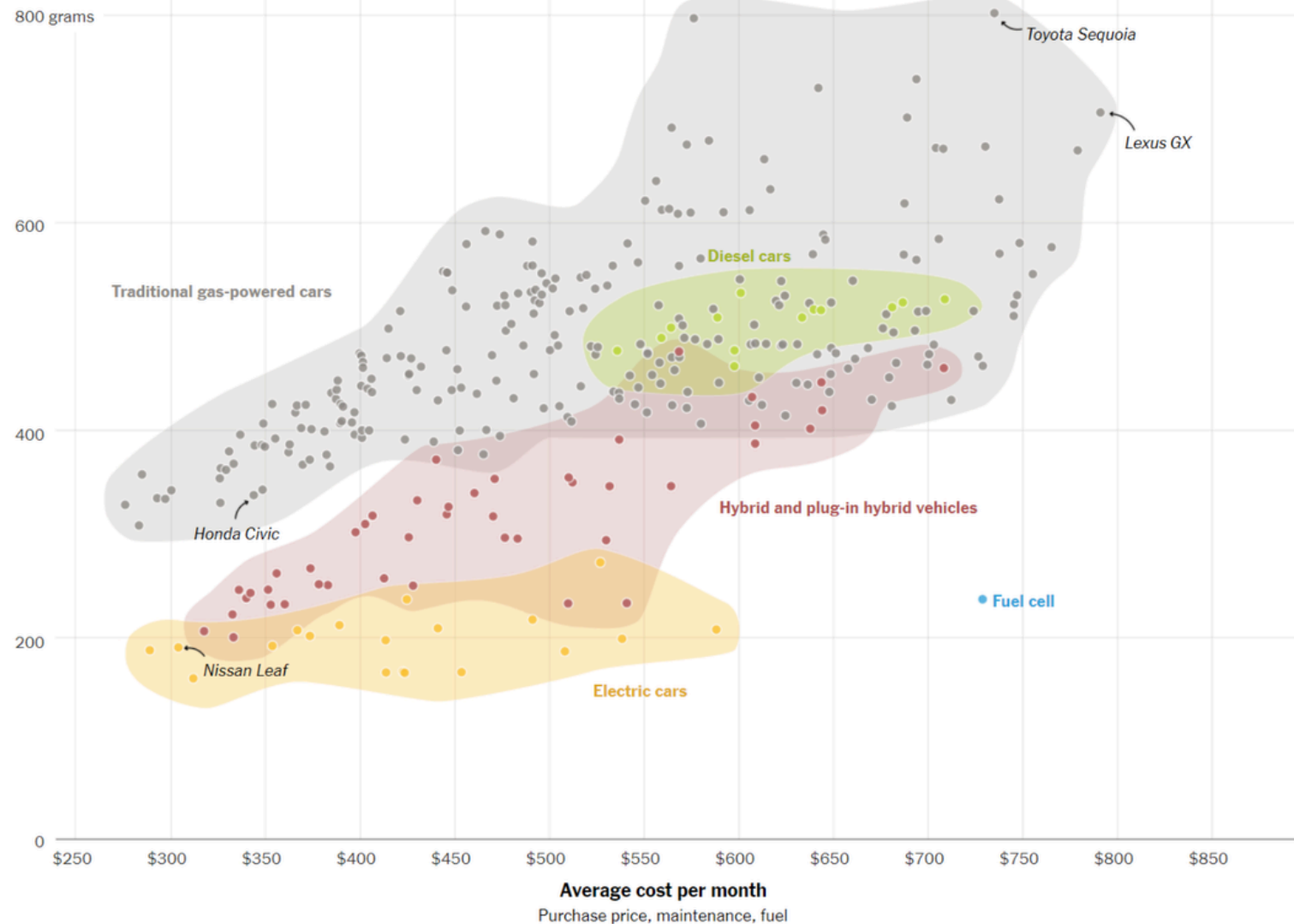
based on the axis
"equality"



https://archive.nytimes.com/www.nytimes.com/interactive/2009/03/01/business/20090301_WageGap.html?_r=0

GROUPS

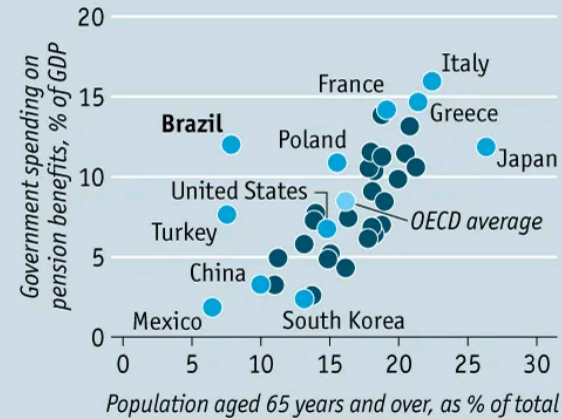
Average carbon dioxide emissions per mile



WHAT
WOULD
YOU
CHANGE
HERE?

Brazil's golden oldie blowout

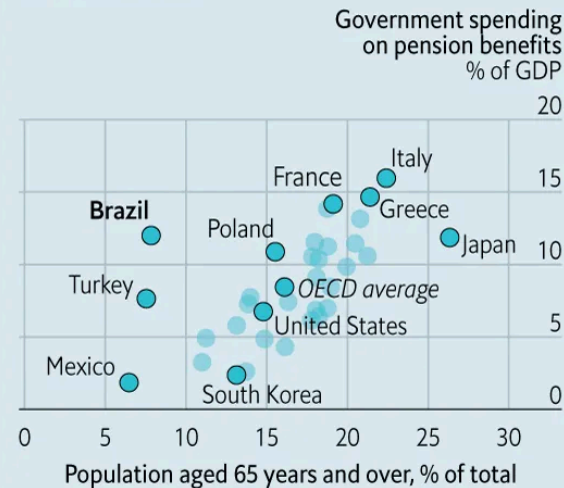
Latest available



Sources: OECD; World Bank; Previdência Social

Brazil's golden oldie blowout

Latest available



Sources: OECD; World Bank; Previdência Social

"In 2016, David Voas and Mark Chaves published an [article in the American Journal of Sociology](#) about how rates of religious commitment in the United States are slowly declining. In 2017, Landon Schnabel and Sean Bock published an [article in Sociological Science](#) responding to this conclusion, arguing that most of the religious decline was among moderate religious respondents—people with very strong religious commitments seemed to be holding steady. Just recently, both teams of authors have published additional comments about this debate ([here](#) and [here](#)), analyzing the same data from the General Social Survey.

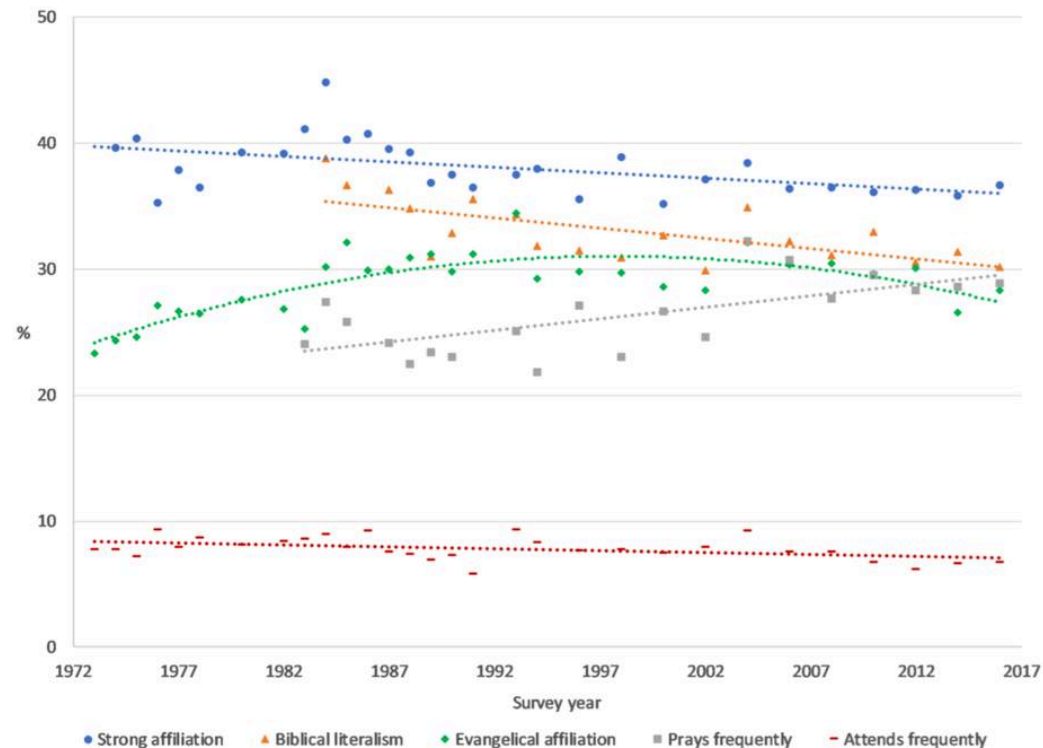


Figure 1: Intense religiosity by survey year (items used by Schnabel and Bock). Source: General Social Survey

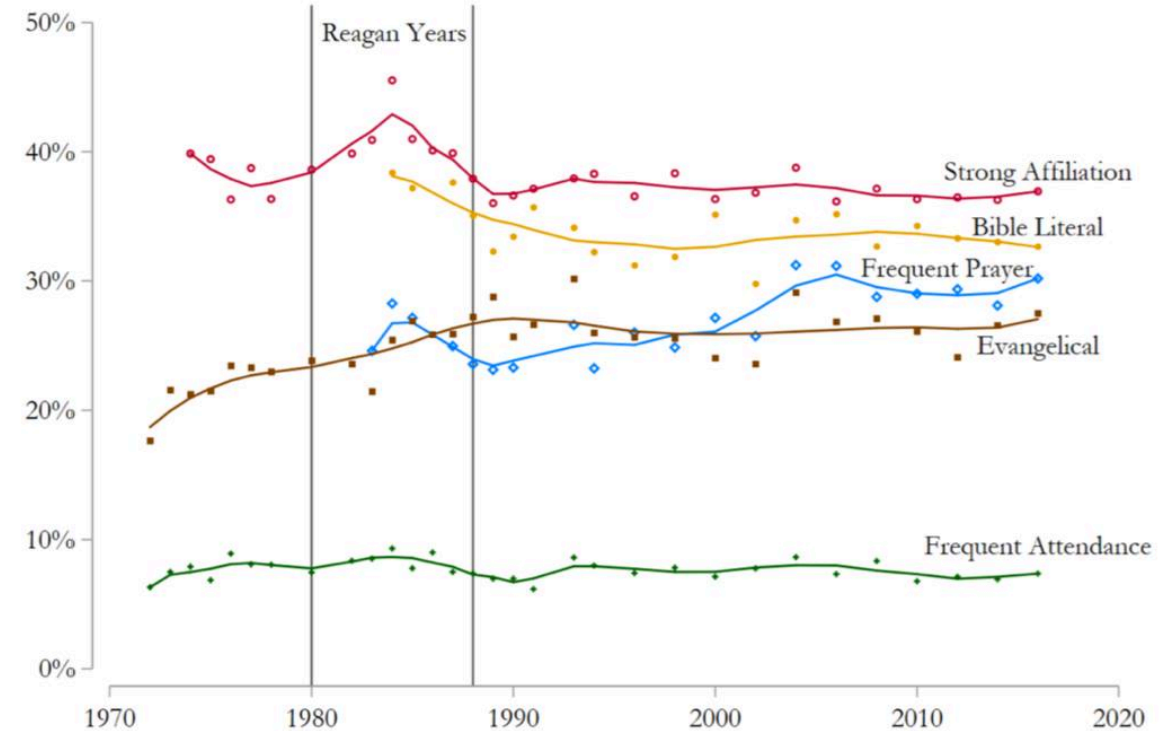


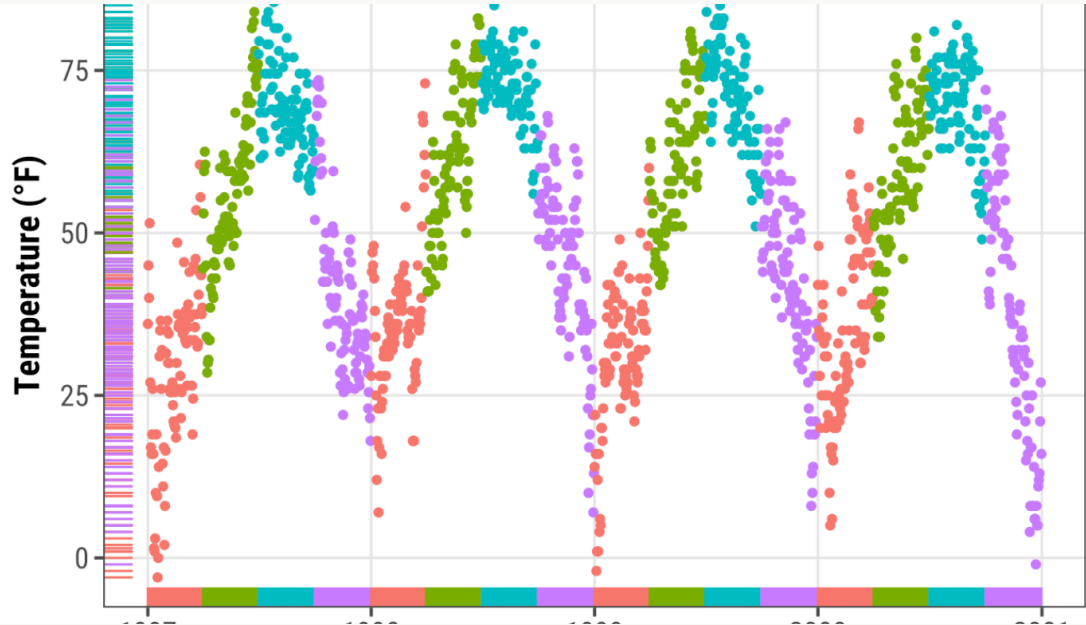
Figure 1: Intense religion by survey year with locally weighted regression lines. General Social Survey

LOESS - WHAT IS IT?
WHY DO WE NEED IT?
WHAT IS A "SPAN"
PARAMETER?

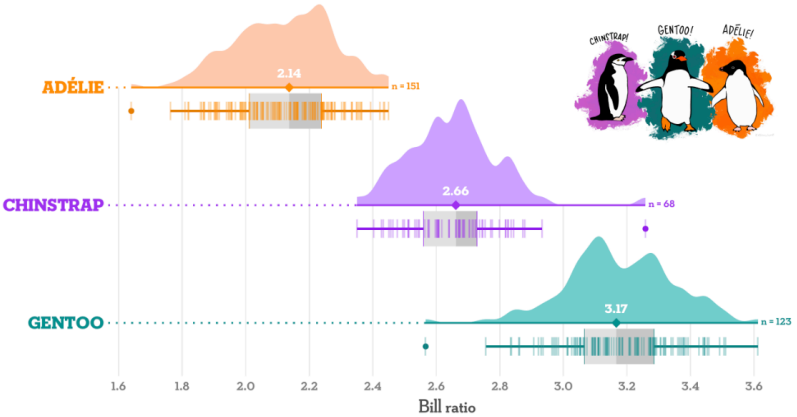


- The default span ≈ 0.75 is a good starting point.
- Larger datasets can support smaller spans (since there's enough data locally for stable estimates).
- Noisy data requires larger spans to smooth over fluctuations.
- The **research purpose** matters: exploratory plots can use smaller spans for flexibility, while confirmatory analysis benefits from smoother, more stable trends.

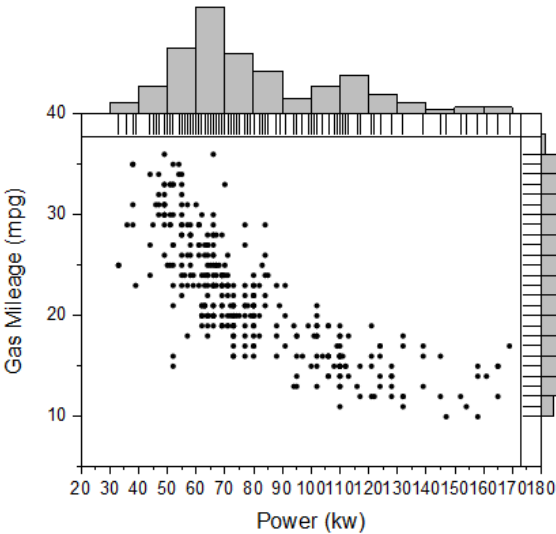
RUGS



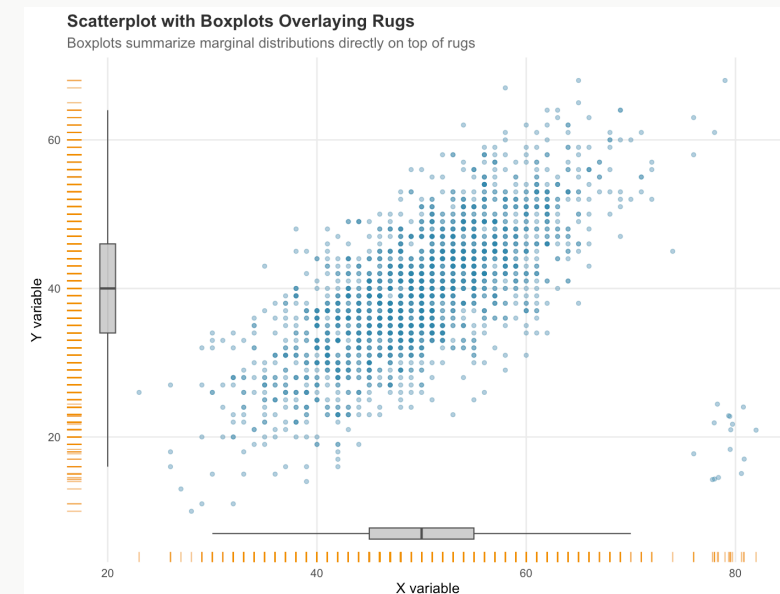
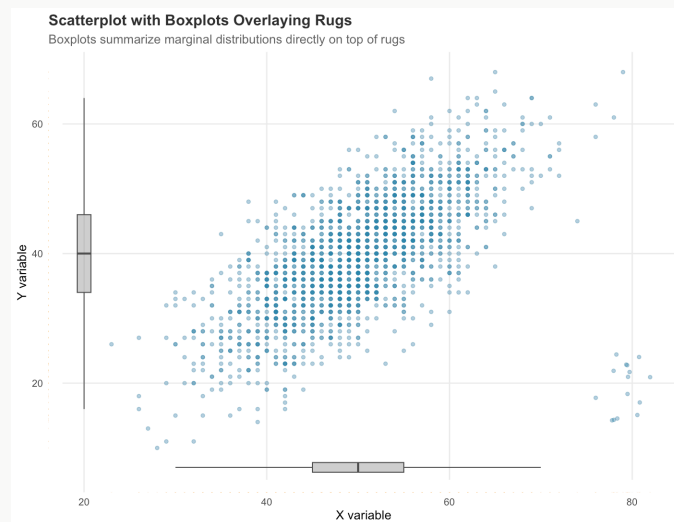
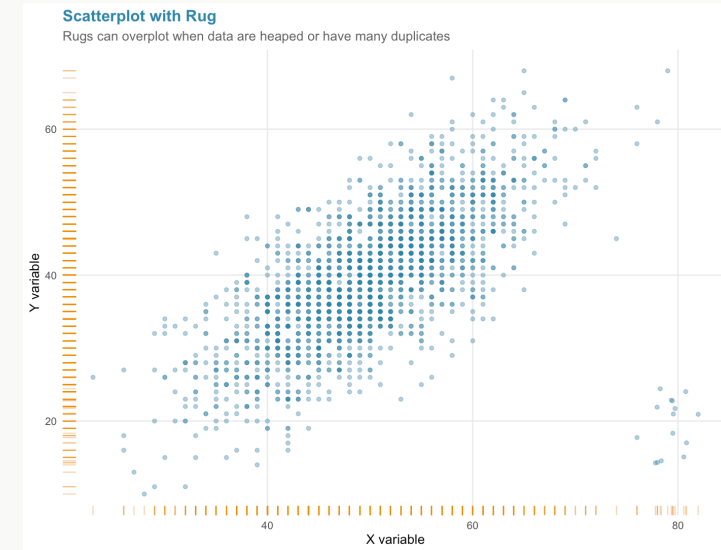
B. Distribution of the bill ratio, estimated as bill length divided by bill depth



Note: In the original data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal (upper) ridge of a bird's bill.
Visualization: Cédric Scherer • Data: Gorman, Williams & Fraser (2014) DOI: 10.1371/journal.pone.0090081 • Illustrations: Allison Horst

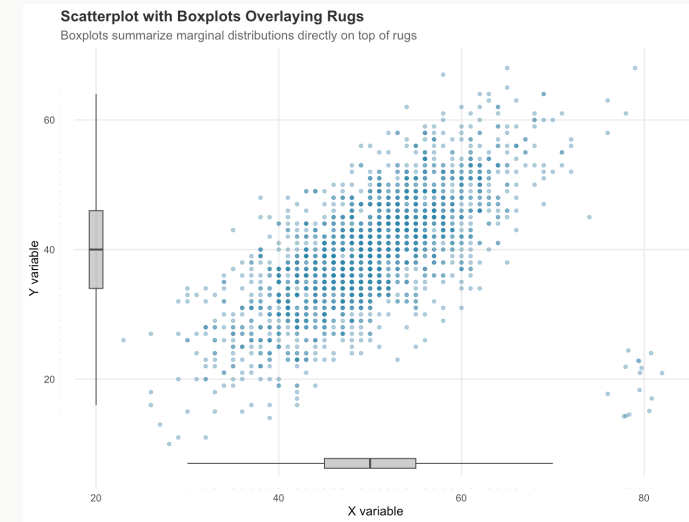


Briana: It was mentioned that you can also use box plots similarly but it may be less desirable since there is less of the data encoded. Is there a situation where box plots might be preferable to rugs despite this?



Max: To me, they seem like a really helpful tool for data comprehension and analysis, but I wonder **how helpful they would be for communicating this data to someone else...**

Basically, is it reasonable to assume that a **layperson could intuit the meaning of this rug visual** on their own, or would the presence of such data just add visual noise and possibly confuse the viewer?



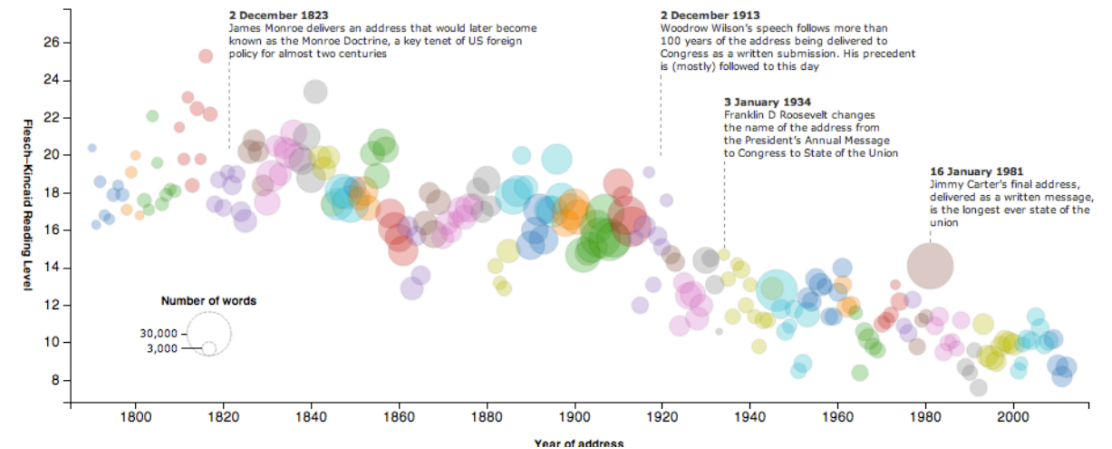
I could not find any Financial Times or NYT or Economist dataviz that used rugs :(

BUBBLES

Annotations are good

Bubble size: radius scales linearly, but the area scales quadratically

State of the Union address decreasing reading level →



Source: The 2011 State of the Union Address

SCATTERPLOTS

Medalists in the Summer Games come from countries with relatively hotter summers and **medalists in the Winter Games** come from countries with relatively colder winters.

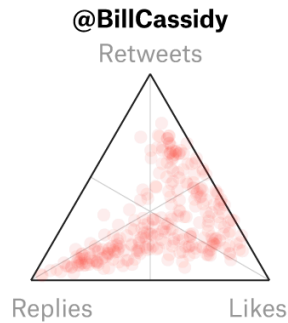


No country warmer than Australia has ever produced a winter medalist at all — including powerhouses like Brazil with 334 medalists in the summer events.
<https://www.datawrapperr.de/blog/summer-winter-olympics-temperature>

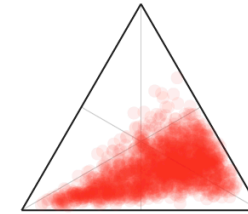
WHAT IF YOU HAVE MORE THAN 2 VARIABLES

The tweets of some high-profile senators

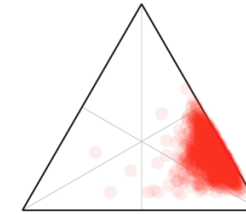
Each senator's 3,200 most recent tweets, excluding tweets with fewer than 10 replies, likes or retweets, through Oct. 19, 2017



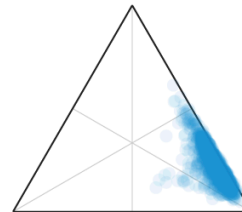
@LindseyGrahamSC



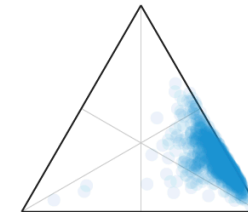
@RandPaul



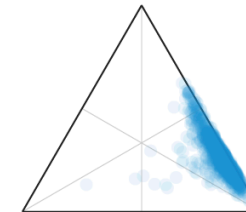
@SenFranken



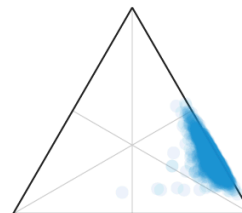
@SenGillibrand



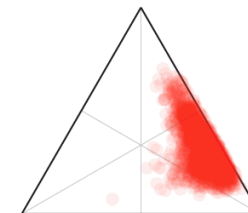
@SenKamalaHarris



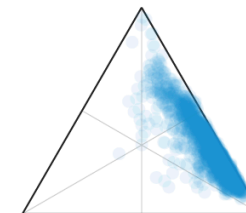
@SenSanders



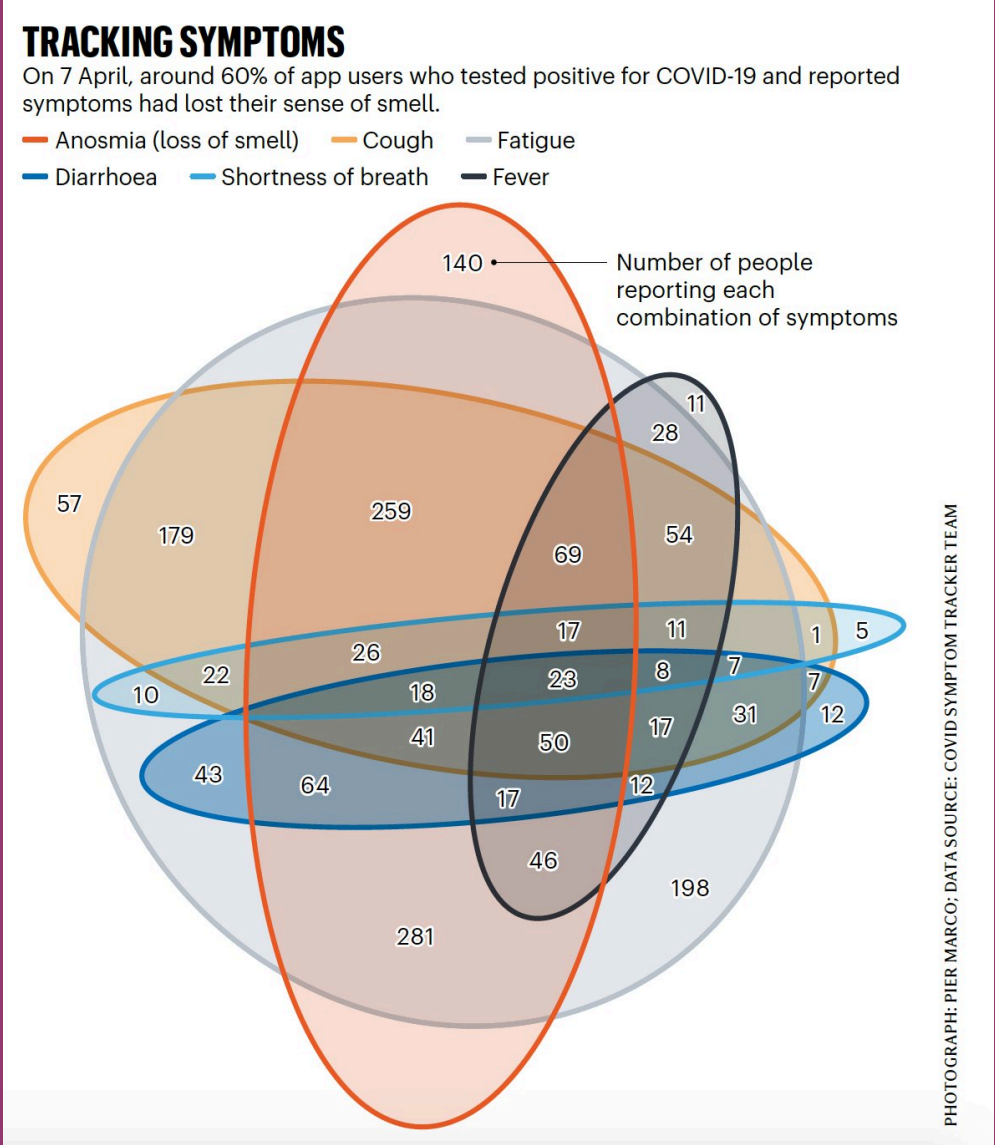
@SenTedCruz



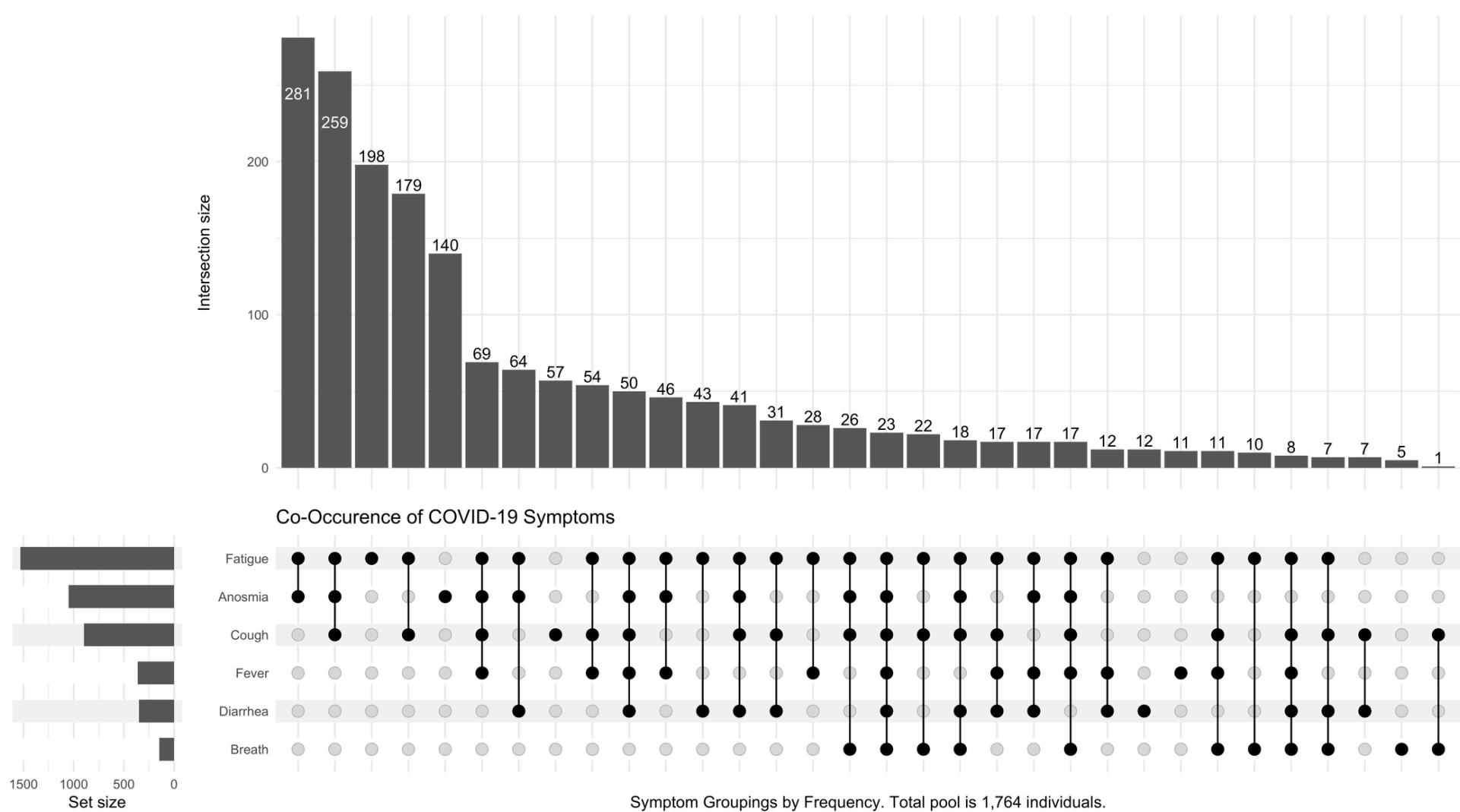
@SenWarren



ONE MORE
INTERESTING
CHART
UPSET PLOT



ONE MORE
INTERESTING
CHART
UPSET PLOT



Symptom Groupings by Frequency. Total pool is 1,764 individuals.

Data: covid.joinzoe.com/us | Graph: @kjhealy